

An LSTM-based Indoor Positioning Method Using Wi-Fi Signals

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ABSTRACT

Recently, Wi-Fi fingerprints are often used for constructing indoor positioning systems. Wi-Fi fingerprint is a vector of Received Signal Strength (RSS) values at a particular location. Radio map is the collection of Wi-Fi fingerprints and their collected location at an area or a building. Positioning systems, mounted on top of the radio map, estimate locations using the information in the radio map. Many Wi-Fi fingerprint-based positioning algorithms have been developed. K-Nearest Neighbor(KNN), probabilistic method, fuzzy logic, neural network, multilayer perceptron are the examples. However, this field has not yet fully benefited from the potential of deep learning approaches. The sequence of Wi-Fi fingerprints implies that the deep recurrent network approaches, especially designed to handle sequential data, can play a vital role to enhance the performance of fingerprint-based positioning systems. In this paper, deep and recurrent approaches are studied rigorously for the improvement of the accuracy of positioning systems. We focus mainly on Long Short-Term Memory (LSTM) networks. An LSTM-based approach was compared with other state of the art approaches. A complete explanation to select the best hyper parameters is presented so that they can be referenced by the researchers in this field. A simple vanilla LSTM architecture is also compared with a stacked LSTM architecture on a Wi-Fi fingerprint dataset.

CCS Concepts

• Networks → Location based

Keywords

Indoor Localization; Sequence Learning; Recurrent Networks

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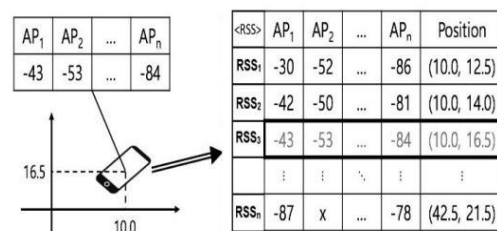
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1. INTRODUCTION

Indoor positioning is the need of the hour and bull's eye for many big companies such as google [1], Microsoft [8] etc. Until now, Wi-Fi dataset has been one of the most promising dataset for fingerprint-based positioning algorithms [7]. Deep Learning has the potential to have a great impact on Wi-Fi indoor positioning system due to its ability to find hidden structure in the input data distribution in order to discover meaningful representations. Deep Learning performs this process often at multiple levels, where higher level learned features are defined in terms of lower level features.

Over the past few years, deep learning is one of the most highly sought after technique in the world of machine learning. This field has been embellished by both academia and industry. Many applications of today's world rely on deep learning and there are many more to come in the future. One application that has still not fully benefited from the deep learning is indoor positioning

Every landmark position has a specific set of received signal strength values (Figure 1). Without deep learning, it is difficult to find any appropriate structure and relationship between RSS values and geolocation coordinates which is important for accurate indoor positioning. By using deep neural networks that are specifically good for handling sequential datasets, accuracy of positioning methods can be improved. Therefore, our motivation behind using LSTM model is to identify whether there is any long-term dependency existing in the Wi-Fi data. This can be identified from the performance of the deep recurrent models. RNN and LSTM architectures are capable of identifying long-term dependencies and uses them for future prediction. For the accurate prediction of points on the radio map, it is useful to consider the previous landmark position. That is what RNN and LSTM aim to do. LSTM cells consume their past experiences of inputs for their next prediction. LSTMs have an ability to add or remove information while going through gates that are a way to decide how much information to carry among the layers.



<RSS>	AP ₁	AP ₂	...	AP _n	Position
RSS ₁	-30	-52	...	-86	(10.0, 12.5)
RSS ₂	-42	-50	...	-81	(10.0, 14.0)
RSS ₃	-43	-53	...	-84	(10.0, 16.5)
...	...	...	...	...	...
RSS _n	-87	x	...	-78	(42.5, 21.5)

Figure 1. Sample of dataset.

In this paper, we propose a scheme to employ LSTM network to learn the mapping between the reference point coordinates and RSS vectors. LSTM is widely used in machine learning in Language Modeling [6], Speech Recognition [14], Image Captioning [18], Handwriting Generation [12] etc. The intuition of applying LSTM is that it can provide a very promising solution to sequence and time series related problems. We not only apply LSTM network for indoor positioning but the focus is on accurate indoor positioning system from sparse RSS vector dataset. We investigate herein the performance of the proposed scenario compared to popular fingerprinting localization algorithms and other state of the art deep learning algorithms. Additionally, we also investigate the impact of various choices of LSTM network design parameters. Work that promotes deep learning usually provides only the performance of the best system, and rarely includes details on how its seemingly optimal parameters were discovered. However, in this paper we provide a full description to decide the best optimal parameters.

2. BACKGROUND

2.1 Indoor Positioning with Deep Learning

There exists a few attempts to use neural networks for Wi-Fi indoor positioning in the past. In [13], a combination of Artificial Neural Network (ANN) and Database Correlation method is used with Wi-Fi fingerprints dataset. M. Stella, M. Russo, and D. Begusic obtained the position by passing Wi-Fi signals through a three layer Perceptron (MLP) [11]. In [15], a Back Propagation Neural Network (BPNN) is employed and they show that Artificial Neural Network is sufficient for Wi-Fi indoor positioning. Another attempt was to use channel state information (CSI), rather than a fingerprint composed of RSS values [16]. They used greedy learning algorithm in the offline phase and reduced the location error.

There exists some indoor positioning approaches in which datasets other than Wi-Fi signals are used with deep learning. In [3], data from both Bluetooth and Wi-Fi is utilized for indoor positioning. They used deep belief networks and the average accuracy achieved by the system is 0.52m. In [5], a methodology based on Recurrent Neural Network is proposed but they use magnetometer data and achieved 1.062m accuracy. This is a strong indication that recurrent networks can be used in indoor positioning with Wi-Fi data also.

2.2 Role of Deep Learning for Sequence Problems

Sequence problems are a special kind of problems that should be handled separately. Classical neural networks called MLPs for short, can be applied to sequence prediction problems [2]. MLPs approximate a mapping function from input sequence to output sequence. However, MLPs are stateless and approximation function is fixed. They generalize all the outputs that are dependent on the input sequence. They are unaware of temporal structure. They consider timestamps as other input features and cannot utilize the order between the inputs as they have no special handling of the timestamps. MLPs have messy scaling criteria. Therefore, they cannot model multiple parallel input sequences. The other major issue that MLPs face to handle sequential problems is that they require fixed size inputs and outputs. It means that size of all the inputs and outputs should be same and they cannot work if the size is not same.

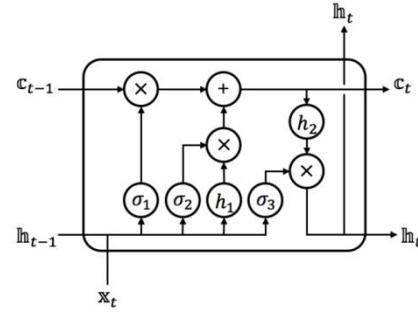


Figure 2. LSTM cell.

RNN is a special type of deep network that handle sequences very well [10]. A simple RNN can be visualized as add-ons of extra loops in MLP architecture. RNN overcomes all the limitations to handle sequence problem that MLP faces. All this discussion sums up to the point that RNN is a quite useful network for sequence problem but on the other hand, it suffers from vanishing gradient problem. To solve this vanishing gradient problem, LSTM was introduced [4]. LSTM is an extension of RNN and gives an even better performance on sequence problems.

In the de facto standard machine learning algorithms used so far for indoor positioning, individual RSS vectors of Wi-Fi signals are treated as statistically independent. However, considering the sequential dependencies of RSS vectors over each other can increase the performance of the system. In this regard, recurrent deep learning models can be very effective. Most prominently, models based on LSTM can enhance the performance. Unlike RNN, LSTM have multiple neural network layers that interact with each other in a special way (figure 2). Explanation of these layers are beyond the scope of this paper.

3. METHODOLOGY

LSTM is implemented in vanilla variant. This is a good starting point to prove the fact that recurrent networks can perform better on Wi-Fi indoor positioning because this a simple and standard LSTM referenced in much work [4].

Vanilla LSTM architecture is simply recurrent because the output gate forms a loop within the network. Input gate and all other gates get the output of the current timestamp and uses it for the next timestamp. It means that the previous network state is not wasted and the previous state is utilized for the next prediction.

Practically LSTM can be implemented in two ways.

1. Bi-directional LSTMs that considers both previous and the next timestamps and it is followed by a layer, which combines their internal states to predict for the current timestamp.
2. Deep forward LSTM that considers only the previous timestamp in multiple layers of recurrent units.

The above two methods of LSTM implementation are significantly different for Wi-Fi indoor positioning. In a forward LSTM design, the previous state is considered to determine the current state. However, in bidirectional LSTM both the previous state and next state are considered to determine the state of current timestamp. For a bidirectional LSTM, it is essential that we know the next timestamp in advance that is why; it is more suitable for offline applications. Since in the forward LSTM design, current timestamp is only dependent on the previous timestamp hence, it

is suitable for real time applications. In this paper, bidirectional LSTM is implemented.

Figure 3 demonstrates the design of our proposed system. Vectors of RSS values are given to the system as input. For wireless signals conforming to the IEEE 802.11 standard, the minimum signal strength that can be received is -100 dBm, and the maximum signal strength that can be received is -10 dBm [17]. However, the training process of LSTM network can be slowed down due to the range of input values, that is from -10 to -100, and it can also be a hindrance in effective learning of the model. Hence, to avoid this situation, input data is first normalized using max-min normalization. After normalization, all the input RSS vectors are in range 0-1.

A special attention is paid to the hyper parameters selection discussed in 3.2. LSTM model is updated until the best hyper parameters are selected and an optimized LSTM model is achieved.

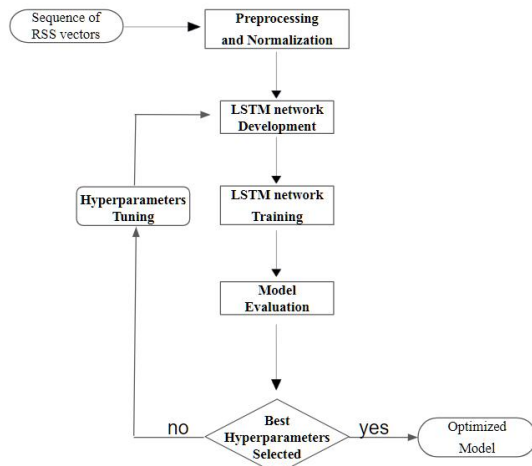


Figure 3. Flowchart of hyper parameters Selection Process.

3.1 Stacked LSTM

In comparison to a shallow network, more functions can be represented with a deep, hierarchical model [9]. That's why, a deeper network is more reliable. Based on this fact, a more deep LSTM version is available that is called Stacked LSTM (Figure 4).

Stacked LSTM architectures with multiple hidden layers can build up progressively higher level of abstractions. They provide a more ideal representation of sequence data, and thus, work more effectively. In Stacked LSTM architecture, current LSTM hidden layer is fed with input from previous LSTM hidden layer. This can substantially increase the performance of neural networks.

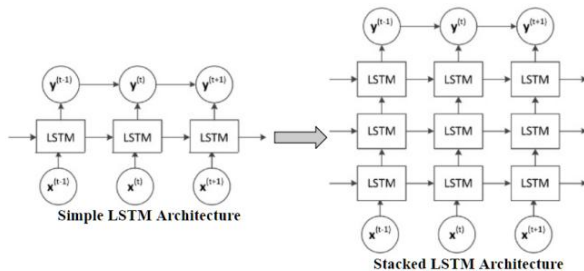


Figure 4. Simple LSTM Vs Stacked LSTM.

The precision and reliability of LSTM model is enhanced by adding enough layers in the model. This leads to requirements of less number of neurons in each layer and reduce the training time. Therefore, a deeper optimized LSTM network is proposed in this study.

3.2 Network Parameters

Training history of our LSTM model is used to diagnose the behavior of the model. LSTMs are stochastic, meaning that each run can give a different diagnostic plot so; performance of the model is plotted against several runs. Reviewing these plots, we decide the best hyper parameters to achieve a better performance.

It is not possible to determine the suitable number of memory cells for the Wi-Fi dataset or LSTM architecture without analysis. A range from 1 to 100 is chosen for the analysis of best suitable number of memory cells in Figure 5. Same is the case with LSTM hidden layers. It is a common convention that deeper the network, more improved is the performance. So, same process is repeated to determine best number of hidden layers in Figure 6.

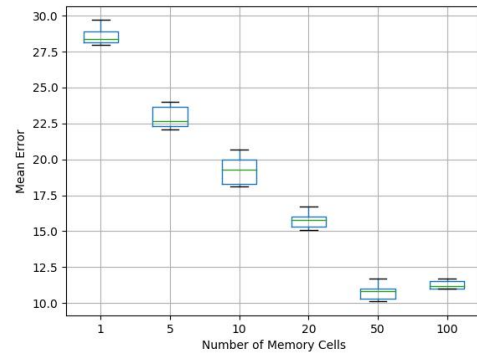


Figure 5. Box plot of memory cells.

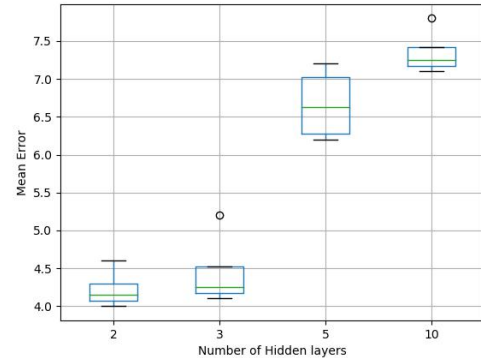


Figure 6. Box plot of hidden layers.

At the end of each batch, weights are updated in accordance with the estimated gradient. This process of weights update depends on the leaning rate that has a large impact on the speed and credibility of the model training. A range of 0.1 to 0.002 is selected to search for the learning rate in Figure 7. Number of epochs are totally dependent on learning rate. For a smaller learning rate, a large number of training epochs are required and vice versa.

Next important parameter in the list is batch size that determines the number of samples in each iteration to update the model weights..

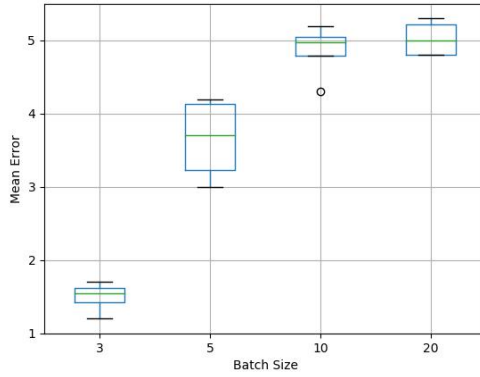


Figure 8. Box Plot of Batch size.

Activation function is responsible for transferring the weighted activation of the neurons of the network. This is determined depending on the scale of input and output layers. Adam algorithm is nice implementation of gradient descent because with this, a customized learning rate is automatically determined for each parameter of the LSTM network. Additionally, best characteristics of AdaGrad and RMSProp methods are also incorporated in this. Hence, Adam optimizer is used as an optimization function.

After considering the above plots against each hyper parameter, finalized set of best hyper parameters is shown in Table 1.

Table 1. Finalized set of best hyper parameters.

Optimizer	Adam
Learning Rate	0.005
Hidden Layers	2
Number of Cells	50
Batch Size	3

4. EVALUATION AND DISCUSSION

4.1 Experimental Data

Wi-Fi fingerprints dataset can be collected in two ways that are

1. Point-by-point
2. Walking Survey dataset.

In point by point data collection, RSS values are collected at each reference point. However, depending on the collection interval, the radio map collection time increases. In fact, collection time depends upon on how many fingerprints are collected in one location. The Walking Survey method has been proposed to improve the drawback that the point by point method takes a long time to collect. This method specifies the path to collect the signal, and the person collects the Wi-Fi fingerprint while walking along the path. The fingerprints allocate positions of the path on the basis of the collected time. Hence, walking survey dataset is collected on 7th floor of N1 building.

4.2 Training and Testing Data

All deep learning models are stochastic. It means that when we train them on a dataset, they choose the initial weights arbitrarily and also rearrange the dataset randomly for each training epoch. This implies that a deep learning model on a single dataset can

give diverse results on multiple runs. Thus, using k fold cross validation is not a good choice.

For the evaluation of our proposed model, dataset is divided separately into training and testing set. The model is fit on training set and evaluated on a separate testing dataset. We collected total 860 training points and 240 testing points.

4.3 Results

In this paper, the performance of KNN is considered as lower level of accuracy. Based on the experiments the value of n is chosen to be 4. In case of MLP, the best results were obtained with 3 layers.

RNN was implemented to assure that recurrent networks can perform better on Wi-Fi dataset. The main baseline model is Vanilla LSTM, which shows 47.8% improvement over KNN and 10.2% improvement over RNN using the full dataset. Then, after updating Vanilla LSTM to 3-Stacked LSTM the accuracy is even improved. 3-Stacked LSTM outperforms KNN by 74.4% and Vanilla LSTM by 18.1%.

Figure 9 shows that LSTM performs better than KNN and RNN even when trained with half dataset.

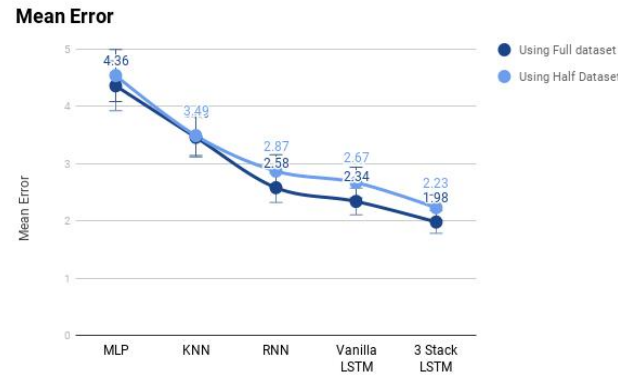


Figure 9. Comparison graph of accuracy using full dataset

It is proven that LSTM shows better accuracy than other models. However, from a practitioner's view the point of interest may not be the higher accuracy of LSTM, but the process for search of best hyper parameters and insights into their suitability for Wi-Fi dataset. All experiments are based on random exploration of parameter space. It is found that performing many iterations for parameter selection is a suitable way to find the best hyper parameters and recommend this approach to practitioners.

It is founded that deep neural networks are very sensitive to hyper parameters and a significant effort is required to explore best suitable parameters. It is recommended not to discard LSTM in preliminary exploration because after selecting the suitable parameters it can perform better. Another finding related to LSTM parameters is that it does not perform better with more than 2 hidden layers for Wi-Fi indoor positioning.

5. CONCLUSION AND FUTURE WORK

So far, indoor positioning relies mostly on the machine learning models that does not involve sequence learning. Inherent to this approach is that it introduces highly appropriate reasons to use LSTM in indoor positioning.

In this paper, performance of state of the art approaches is explored for indoor positioning. A detailed explanation on how to

train LSTM effectively and select the best hyper parameters is presented. It is successfully proven from the evaluation that bi-directional LSTM outperforms other machine learning approaches by a considerable margin.

Labeled Dataset collection is a laborious part of indoor positioning. Based on the conclusion of this paper, LSTM can be used for indoor positioning in semi-supervised scenario also.

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